AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 31 · TEACHER GUIDE

Design the Decision Process

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach decisions through state and evidence.

Model the full activity search, inspect premature stopping and distinguish the kit example before the independent room flow.

Keep one habit: Look for every final path, unchanged inputs and finite forward progress.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Design a decision **process** that covers valid, invalid and no-match requests

Keep Week 30's requirements separate from the algorithm that implements them. The manual **baseline** marked all rows; this design may stop at the first fit while preserving the required **output**.

Check more than final labels: pupils must explain both conditions, **validation** and progress through the finite catalogue. This is an explicit-rule recommender, not a model fitted from examples.

By the end, I can design a complete **decision flow** with **validation**, search and stopping.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- The unchanged B1 to B4 catalogue and Week 30 requirements
- Four **candidate** cards and a marker for the current row
- Paper for a **process** drawing and a trace with **candidate**, checks and next step

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

Look for: Turn the project outcome into precise **input** and **output** requirements

Reference example, not the only acceptable wording: Time required must be no more than available time; when quiet is required the activity must be quiet. Choose the first compatible activity in B1 to B4 order.

No means quiet is not required, not forbidden, so quiet B1 is permitted if 10 minutes fits.

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a system’s capability, the correctness of its **output** and permission to take an action

Reference example, not the only acceptable wording: The button establishes a capability. The fact check establishes factual support for A at that time.

The approval permits the specified classroom-board action within its limits. Public posting is outside that permission even though the button exists and A is factually checked.

Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish retrieving a record, calculating from **input**s and generating wording

Reference example, not the only acceptable wording: Room: retrieval, checked against the right source N2 and event. Remaining places: calculation, checked as 6 × 4 − 3 = 21 with correct **input**s.

Title: generation, checked for the requested style and unsupported factual details. An application can combine them; the labels describe operations rather than exclusive types of application.

Blank time requires clarification because an **input** is missing. Zero is valid within 0 to 30, but no activity in the catalogue fits it, so the correct result is no match.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Choose Inputs and Outputs

Humans Set Boundaries

Generate, Retrieve, or Calculate?

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Rule: if paper then P, otherwise M. **Input**s are paper, metal, glass and unknown. Repair the rule for paper and metal trays without inventing a glass tray. Trace all four **input**s.

Concept to explain

Rules or learned patterns must connect **input**s to **output**s transparently.

Term	Meaning
decision flow	ordered path connecting inputs to outputs
baseline	simple method used for comparison
branch	path chosen by a condition
explanation	traceable reason for an output

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; **explanation** 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach complete branches and termination with visible state

> Preserve the activity contract, then distinguish the kit and independent room examples.

Read the unchanged activity contract

A complete **process** says what happens for every kind of request in the task. It must handle a **fitting** activity, no **fitting** activity and unusable **input**. We will design a paper algorithm before building the prototype.

Its decisions use the requirements from Week 30, so a different drawing should not silently change what counts as correct.

Keep the same catalogue order: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These are fixed paper-model values with materials assumed available. Valid **input** is a whole number of minutes from 0 to 30 and Quiet required Yes or No.

A **candidate** fits when its required minutes are no more than the available minutes and either quiet is not required or the activity is quiet. Both parts must hold.

The **output** is the first **fitting** activity in catalogue order, with a reason, or a clear no-match result. Invalid or missing **input** produces a clarification request. None of these **output**s books materials or publishes a message.

Model validation and ordered search

Begin by checking the request. If minutes are missing, outside 0 to 30 or not a whole number, ask for a whole number in that range. If the quiet field is not Yes or No, ask for one of those values.

State the fields needing correction and stop this run. When a corrected request arrives, validate it as a new run rather than continuing with old **candidate state**.

For a valid request, start at B1. Check time and the quiet condition for the current **candidate**. If both pass, **output** that **candidate** and stop. If either fails, move forward exactly one row.

After B4 fails, there is no next row, so **output** no match and stop. Never jump back to B1 within the same search.

Trace candidate changes and a finite stopping argument

Stage for 9 minutes and Quiet required No	Evidence	Next state or output
Validate	9 is a whole number in range; No is valid	Current candidate B1
Check B1	10 minutes is more than 9	Move to B2
Check B2	8 is no more than 9; quiet is not required	Suggest B2 and stop

The current **candidate** changes from B1 to B2. The request values stay the same. After B2 passes, the **process** stops: it does not replace B2 with a later **fitting** row.

This early stop preserves the first-match rule and gives the same required suggestion as the manual **baseline**, which marked every row before selecting the first fit.

Termination needs a reason. There are only four rows, each failed check moves to the next row, and no row is revisited. A valid request therefore examines at most four **candidate**s. A success can stop sooner.

Invalid **input** stops before the search. Merely drawing an arrow around a loop is not enough; the **state** must make progress toward an end.

Introduce the separate kit records

A fictional worksheet routes craft kits by required glue and number of scissors.

K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4.

Request: glue required Yes, scissors needed 2. Return first kit meeting both requirements.

Compare with the activity recommender's four paths

AI ACADEMY / AI-1/31 · 03 · INSPECT THE LESSON MODEL

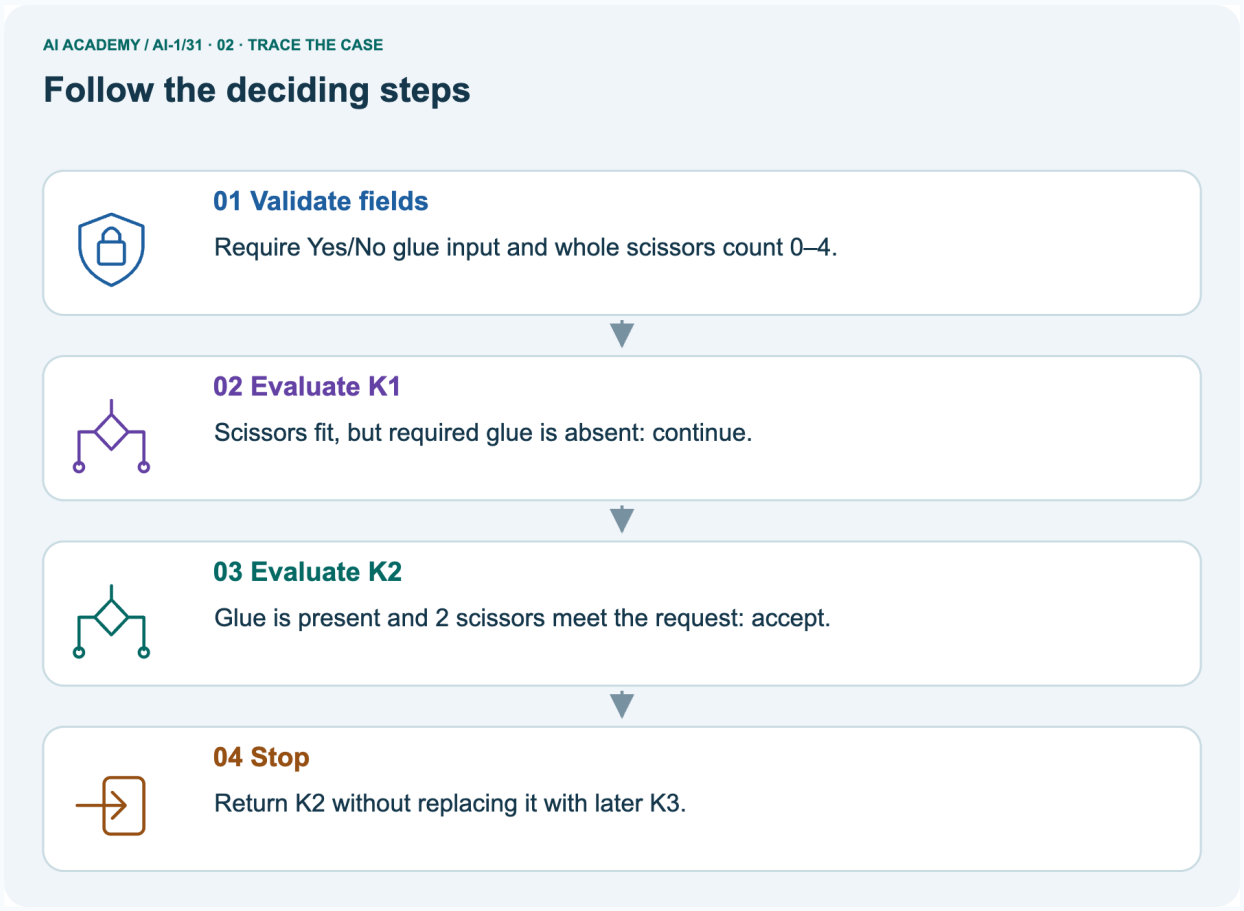
Four paths through the recommender

Condition reached	Next action	Where the path ends
Input missing or invalid	Ask for corrected input	Stop this run
Candidate fails a check	Move to the next row	Continue search
Candidate passes both	Suggest it with a reason	Stop this run
All rows checked and failed	Report no match	Stop this run

Failing one candidate continues the search. No match is justified only after the valid request has no fitting candidate.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the first passing kit and stop



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate fields	Require Yes/No glue input and whole scissors count 0–4.
2. Evaluate K1	Scissors fit, but required glue is absent: continue.
3. Evaluate K2	Glue is present and 2 scissors meet the request: accept.
4. Stop	Return K2 without replacing it with later K3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

Separate flow correctness from inventory truth

A correct flow depends on accurate catalogue entries and the declared requirements; the picture does not verify inventory.

Increase the scissors requirement

Change: Increase scissors needed to 3.

Prediction: K3 is selected.

Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Elicit each next state from the picture

Use the diagram and verbal **explanation** together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

> **1. Input — information entering**
>
> **INPUT** approved **input** fields

> **2. Process — the deciding step**
>
> **PROCESS** smallest rule or learned **decision flow** with visible **branch**es

> **3. Output — the result**
>
> **OUTPUT** suggested bin, not sure, and reason

CORE IDEA Rules or learned patterns must connect **input**s to **output**s transparently.
</details>

UNDER THE HOOD A decision **process** operationalizes the interface contract. Simplicity improves testability and **explanation**, but oversimplification can erase relevant distinctions.

Learned components need explicit data, threshold, and fallback documentation.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE A transparent sorter first checks item type, then one disambiguating feature, then sends unclear items to review.

STEP 1 — OBSERVE approved **input** fields

STEP 2 — TRACE smallest rule or learned **decision flow** with visible **branch**es

STEP 3 — CONCLUDE suggested bin, not sure, and reason

MODEL REASONING Start with a **baseline** whose **branch**es can be read aloud. Each **input** must affect a decision or be removed. Add not-sure and review paths, then explain when a learned component would earn its added complexity.

DO NOT OVERCLAIM Transparent does not mean correct; the flow still needs testing.

What new **evidence** could change this answer?

Concrete teaching case

A rule sends paper to P and everything else to M. **Input**s include paper, metal and glass.

Trace glass and repair the rule.

Exact explanation to model

Glass incorrectly goes to M. Add an explicit metal check and a glass or review **branch**. Otherwise includes all remaining cases.

Explain the added worked example

Key terms

Term	Meaning
Validation	Checking an input against the agreed rules before using it
Branch	A path chosen according to a condition

Term	Meaning
Candidate	An option currently being checked
State	Information that records the process's current position or values
Termination	Reaching a point where the process stops

Model the reasoning aloud

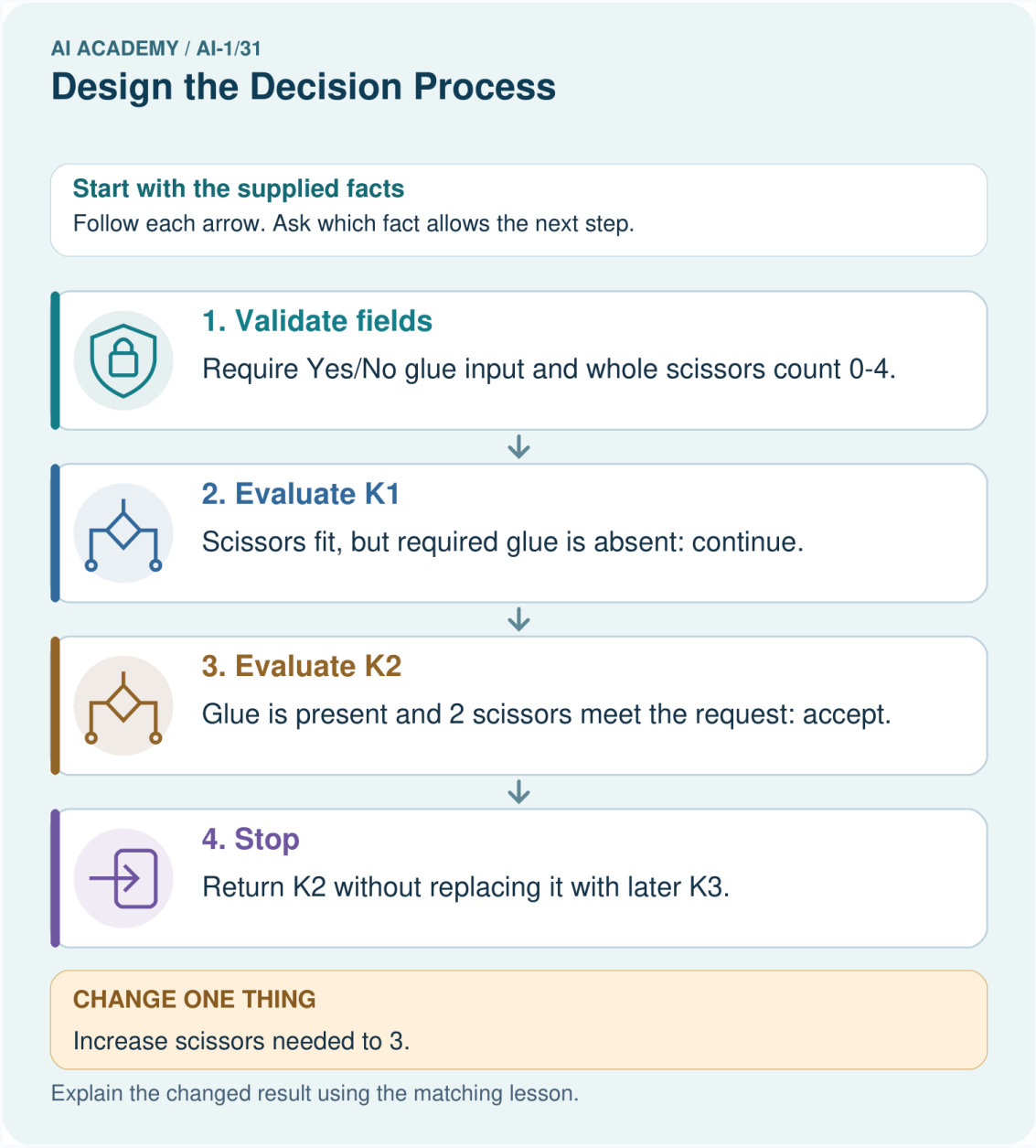
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Design the Decision Process

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Trace 9 minutes with Quiet required No. Record the current **candidate**, time result, quiet condition and next step until the **process** stops.

Explain why the request values should not change as the **candidate** advances.

Expected reasoning and fading help

Teacher check: B1 fails time because 10 is greater than 9; quiet is unrestricted. Advance to B2. B2 passes time because 8 is no more than 9 and quiet is not required, so suggest B2 and stop.

Candidate state changes to search the catalogue; changing the request values would solve a different task.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for Designing a complete decision **process**. Proposed **explanation**: “One failed row proves no match”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Later rows may fit; continue until a fit or catalogue exhaustion.

Collect **explanation**s, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group **explanation** while others solve a more demanding variation.

Use the pictorial check

Where does the first original rejection occur, and why is K3 not originally selected?

K1 fails glue. K2 meets both conditions, so the first-match flow stops before considering K3.

Changed-case reasoning: K3 is selected. K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Collect each learner's **explanation** before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an **explanation**.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ **Earlier wording, alternatives and picture transcripts**

Demonstrate and guide practice

Board sequence

Write the given facts: A rule sends paper to P and everything else to M. **Input**s include paper, metal and glass.

Write the question to explain: Trace glass and repair the rule.

Reveal after **prediction**s: Glass incorrectly goes to M. Add an explicit metal check and a glass or review **branch**. Otherwise includes all remaining cases.

Use the earlier changed question as optional later practice. Reserve Design a complete room decision **process** for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

9 Yes: B1 takes 10, too long; B2 takes 8 but is not quiet; B3 takes 12, too long; B4 takes 15 and is not quiet. After all four fail, **output** no match.

Rechecking a failed row without advancing could repeat forever with unchanged **state**. 31 No is outside the accepted time range, so request a whole time from 0 to 30 and stop before searching.

Exit: B4 is compatible for 15 No, but violates the first-match rule; the required **output** is B1.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve blank input versus a valid zero and no match.
7 to 20 minutes	Identify all endings and branches using the diagram; complete W1.
20 to 33 minutes	Trace 9 No and complete W2 with visible candidate state.
33 to 45 minutes	Complete W3 and contrast both conditions with either condition.
Session 2 start 0 to 8 minutes	Retrieve the first-match rule and the meaning of No.

Stage	Teacher action
8 to 23 minutes	Complete the guided no-match trace and demonstrate a loop without progress.
23 to 38 minutes	Collect W4 independently; ask a partner to follow the written process literally.
38 to 45 minutes	Complete W5 and the exit; save the versioned design for the prototype.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

For the supplied flow audit, F returns B2 for (9,No), no match for (5,Yes), B1 for (11,No), and clarification for (blank,Yes). G wrongly reports no match on (9,No) after rejecting B1; it agrees on the other supplied outputs but that does not repair its incomplete search. Revise failure to advance one row, preserving both checks and first-match stopping. There are four candidates; each rejection advances and no row is revisited. Invalid input stops before search. Keep the original G trace and revised trace distinct. Reserve Design a complete room decision process for an independent first attempt.

Run and record

1. Complete the task: Create and trace the smallest useful **decision flow**.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

One failed row proves no match: Later rows may fit; continue until a fit or catalogue exhaustion.

Passing either condition is sufficient: The time and quiet conditions must both hold.

Any compatible answer satisfies the task: The first-in-catalogue tie rule is also a requirement.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Design a complete room decision process for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO REPAIR Only successful labels need a **branch**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A loop stops because it should: Show a **state** change that advances toward a finite end.

Offer an oral **explanation**, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A new paper catalogue lists C1 first, capacity 4, quiet Yes; C2 second, capacity 8, quiet No. Valid requests give a whole group size from 1 to 10 and Quiet required Yes or No.

A room fits if capacity is at least group size and the quiet condition is met; choose the first fit. Write a complete **process** with **validation**, search, **state** change and stopping rules.

Trace 6 people No and explain the outcome for a blank quiet field. No booking takes place.

Validate group size and Yes or No; request clarification and stop if invalid. Start C1; check capacity and quiet. If both pass, suggest and stop; otherwise advance to C2. If C2 also fails, report no match and stop.

For 6 No, C1 fails capacity, C2 passes capacity with quiet unrestricted, so suggest C2. Blank quiet requires clarification before searching. Each failed row advances, so at most two **candidate**s are checked.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Describe all possible kinds of ending

Name the three kinds of final **output** in the activity **process** and the condition that leads to each. Explain why a **candidate** failing its time check is not automatically a final no-match result.

Clarification follows invalid or missing **input**. A suggestion follows the first **candidate** passing both compatibility conditions. No match follows a valid request after all **candidate**s fail.

A single failed **candidate** does not settle whether later **candidate**s fit, so the **process** must advance.

W2 Track the current candidate

Trace 9 minutes with Quiet required No. Record the current **candidate**, time result, quiet condition and next step until the **process** stops. Explain why the request values should not change as the **candidate** advances.

B1 fails time because 10 is greater than 9; quiet is unrestricted. Advance to B2. B2 passes time because 8 is no more than 9 and quiet is not required, so suggest B2 and stop.

Candidate state changes to search the catalogue; changing the request values would solve a different task.

W3 Repair an incorrect branch

A design says “If the activity is quiet OR its time fits, suggest it.” For 5 minutes with Quiet required Yes it selects B1. Identify the failed requirement and replace the condition with a correct one.

Give the correct result for this request.

B1 is quiet but requires 10 minutes, exceeding 5. Require time to fit AND the quiet condition to hold. The quiet condition is either quiet not required or a quiet activity.

No catalogue activity fits 5 minutes, so the correct **output** is no match after checking all **candidate**s.

W5 Preserve the first compatible option

For the activity catalogue, 15 minutes with Quiet required No allows several activities. Explain why returning the last compatible row breaks the agreed rule even though it fits.

Describe a stopping rule that preserves the first match, and one way to check that a loop cannot repeat forever.

All four activities fit, but the tie rule requires B1 first. Returning B4 meets compatibility but not the full specification. Stop immediately after the first passing **candidate**.

Check that every failure advances exactly one row, no row is revisited and exhaustion of B4 ends with no match.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Glass incorrectly goes to M. Add an explicit metal check and a glass or review **branch**. Otherwise includes all remaining cases.

Additional transfer practice question

Write a decision **process**: a clear pencil goes to the pencil tray, a clear book to the shelf, and an unknown object to review. Trace a lunch box.

Reasoned practice answer

Check for a pencil, then a book; if neither is supported, request review. A lunch box follows the review **branch**. The fallback is part of the **process**.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Only successful labels need a **branch**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Use if paper then P; else if metal then M; else review. Results are P, M, review, review. The catch-all must match the available actions. Never label all non-paper objects as metal merely because there are two trays.

Explanations for the additional workbook examples

W1 Describe all possible kinds of ending

W2 Track the current candidate

W3 Repair an incorrect branch

W4 Design a complete room decision process

A new paper catalogue lists C1 first, capacity 4, quiet Yes; C2 second, capacity 8, quiet No. Valid requests give a whole group size from 1 to 10 and Quiet required Yes or No.

A room fits if capacity is at least group size and the quiet condition is met; choose the first fit. Write a complete **process** with **validation**, search, **state** change and stopping rules.

Trace 6 people No and explain the outcome for a blank quiet field. No booking takes place.

Validate group size and Yes or No; request clarification and stop if invalid. Start C1; check capacity and quiet. If both pass, suggest and stop; otherwise advance to C2. If C2 also fails, report no match and stop.

For 6 No, C1 fails capacity, C2 passes capacity with quiet unrestricted, so suggest C2. Blank quiet requires clarification before searching. Each failed row advances, so at most two **candidate**s are checked.

W5 Preserve the first compatible option

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Complete the sorter v1 **decision flow** and annotate every **branch** with the **evidence** it uses.

Keep the decision, **input**, result, **explanation** and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate.

Support while preserving the reasoning goal

Use physical cards, spoken **explanation**s and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Move a marker down the **candidate** cards while leaving the request card fixed. Use separate time and quiet ticks before combining them.

Extension: compare the number of **candidate**s examined by the early-stop **process** and the full manual checklist without claiming a measured time saving.

W4 must contain **input validation**, both room checks, first-match stopping, advancement, exhaustion and the C2 trace for 6 No. Blank quiet must produce clarification.

After support change the request to 6 Yes: both rooms fail different conditions, so no match after two checks. Require an **explanation** that this is different from invalid **input**.

Plan the delayed check

Suggested return points in this level: ai-1/32, ai-1/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 32: Build a Paper Prototype. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 32 builds a paper prototype from the decision **process** and records each step as a trace.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Design a decision **process** that covers valid, invalid and no-match requests

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Trace 9 minutes with Quiet required No. Record the current **candidate**, time result, quiet condition and next step until the **process** stops.

Explain why the request values should not change as the **candidate** advances.

Reasoned practice answer: B1 fails time because 10 is greater than 9; quiet is unrestricted. Advance to B2. B2 passes time because 8 is no more than 9 and quiet is not required, so suggest B2 and stop.

Candidate state changes to search the catalogue; changing the request values would solve a different task.

Check every learner before discussion: Use the worked case for Designing a complete decision **process**. Proposed **explanation**: “One failed row proves no match”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Later rows may fit; continue until a fit or catalogue exhaustion.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken **explanation** or drawing; retain the reasoning goal.

Extension: Increase scissors needed to 3. Require a **prediction** and a reason before comparing. Expected result: K3 is selected. Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's **explanation** and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-**explanation** links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess **explanation**, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an **explanation**. The **branch**-boundary and indexing lessons deliberately include failure cases.
</details>

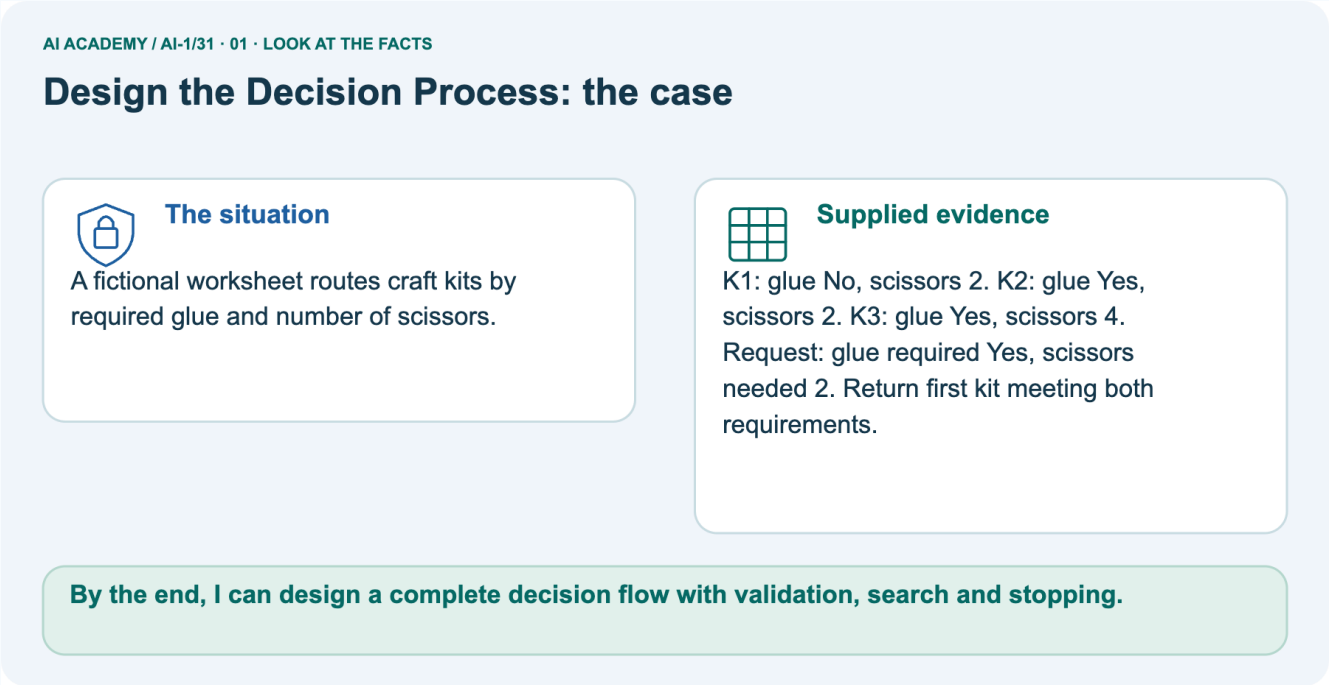
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Design the Decision Process: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

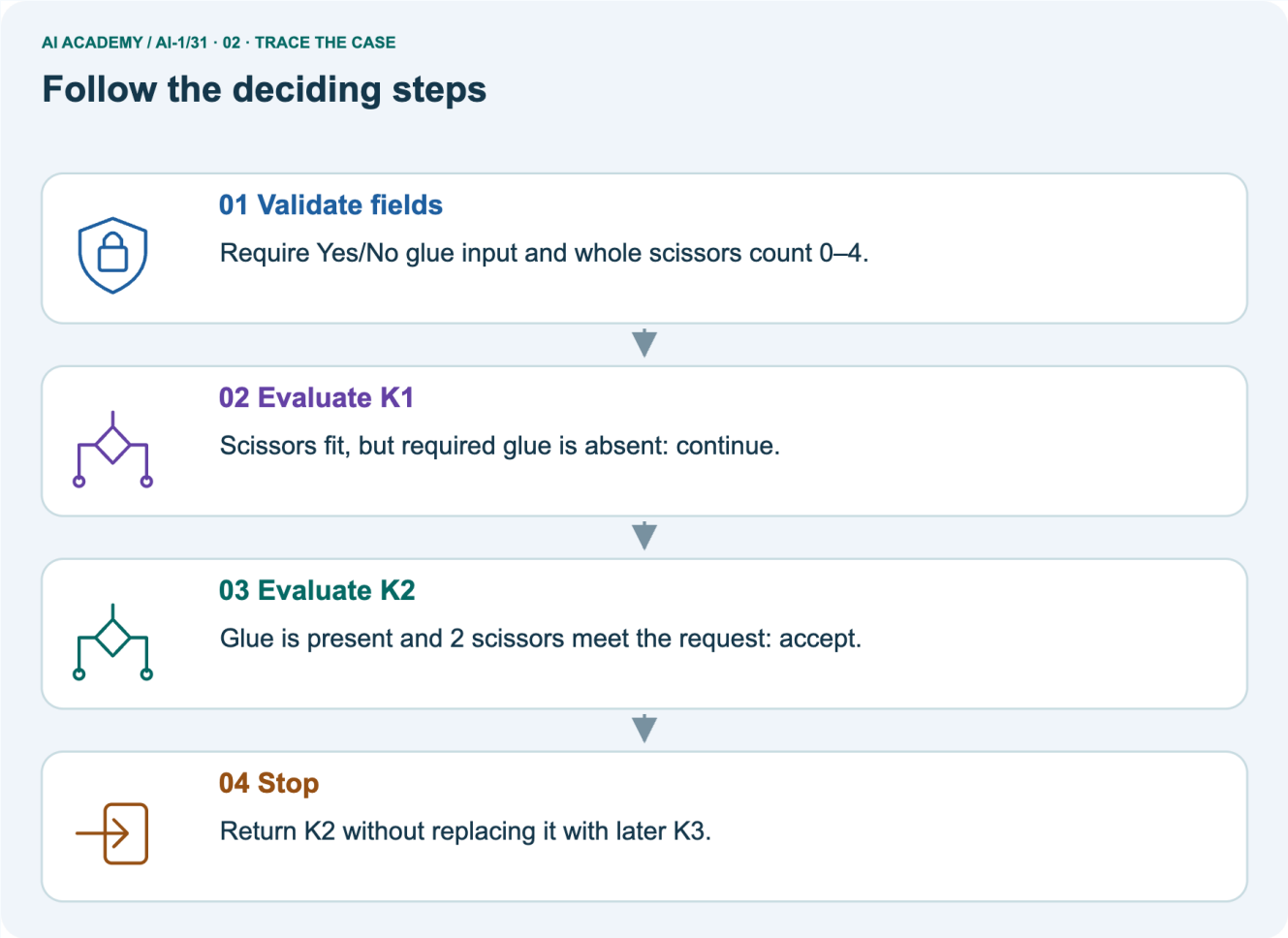
A fictional worksheet routes craft kits by required glue and number of scissors. Supplied facts: K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4. Request: glue required Yes, scissors needed 2.

Return first kit meeting both requirements.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



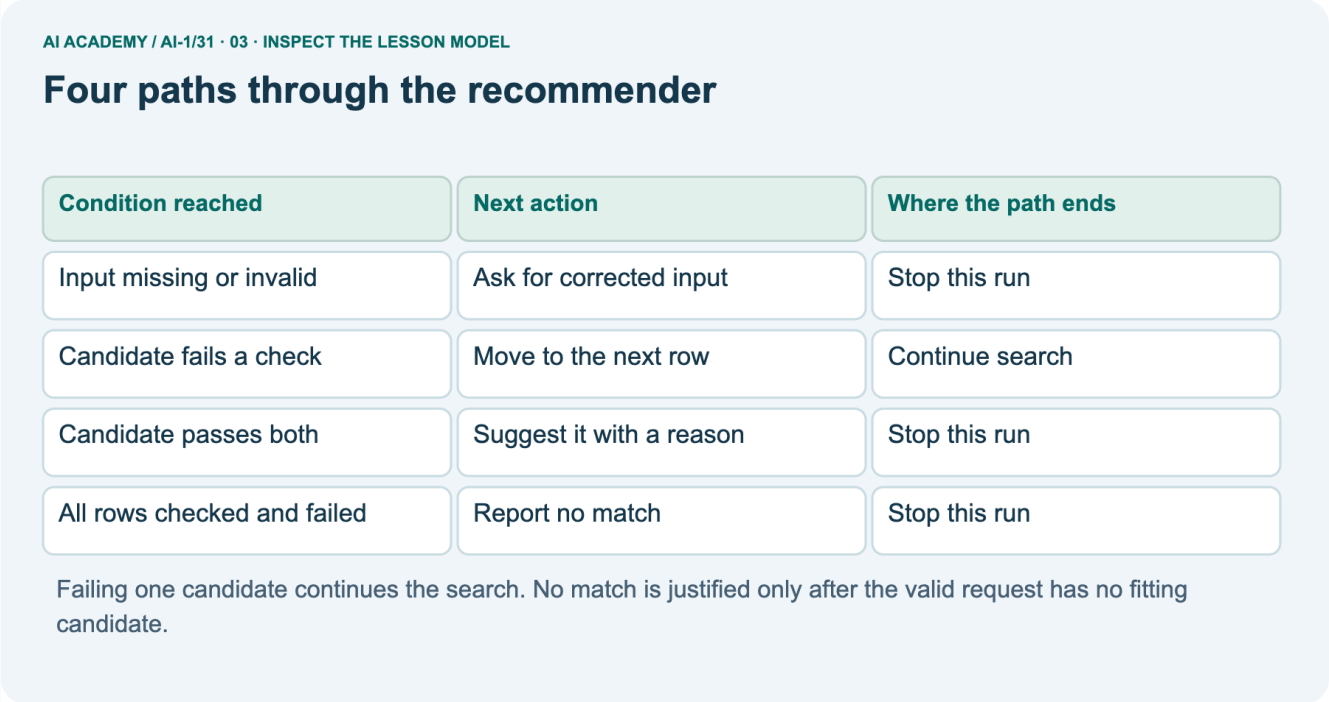
Follow the labelled picture with the matching explanation. Open the vector diagram

Validate fields: Require Yes/No glue **input** and whole scissors count 0–4. → Evaluate K1: Scissors fit, but required glue is absent: continue. → Evaluate K2: Glue is present and 2 scissors meet the request: accept. → Stop: Return K2 without replacing it with later K3.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Four paths through the recommender



AI ACADEMY / AI-1/31 · 03 · INSPECT THE LESSON MODEL

Four paths through the recommender

Condition reached	Next action	Where the path ends
Input missing or invalid	Ask for corrected input	Stop this run
Candidate fails a check	Move to the next row	Continue search
Candidate passes both	Suggest it with a reason	Stop this run
All rows checked and failed	Report no match	Stop this run

Failing one candidate continues the search. No match is justified only after the valid request has no fitting candidate.

Follow the labelled picture with the matching explanation. Open the vector diagram

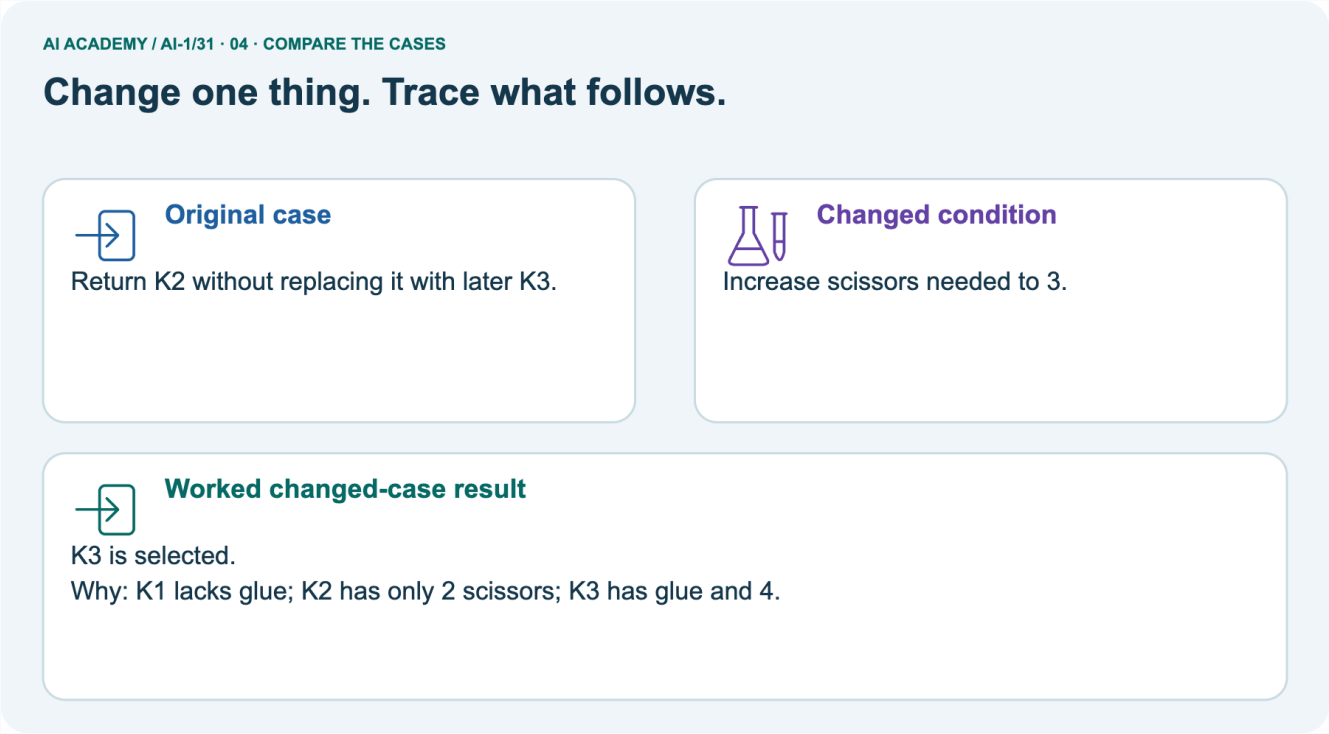
Four paths through the recommender. **Input** missing or invalid | Ask for corrected **input** | Stop this run. **Candidate** fails a check | Move to the next row | Continue search.

Candidate passes both | Suggest it with a reason | Stop this run. All rows checked and failed | Report no match | Stop this run

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

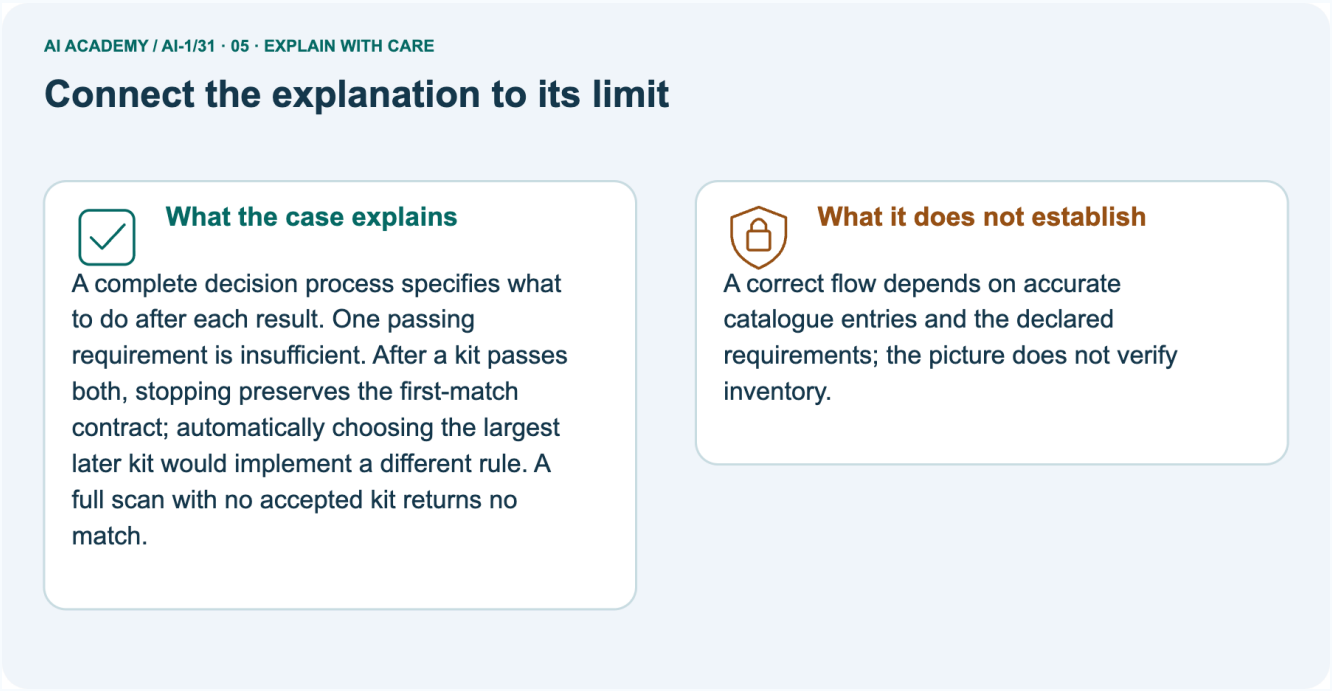
Original case: Return K2 without replacing it with later K3. Changed condition: Increase scissors needed to 3. Expected result: K3 is selected. Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

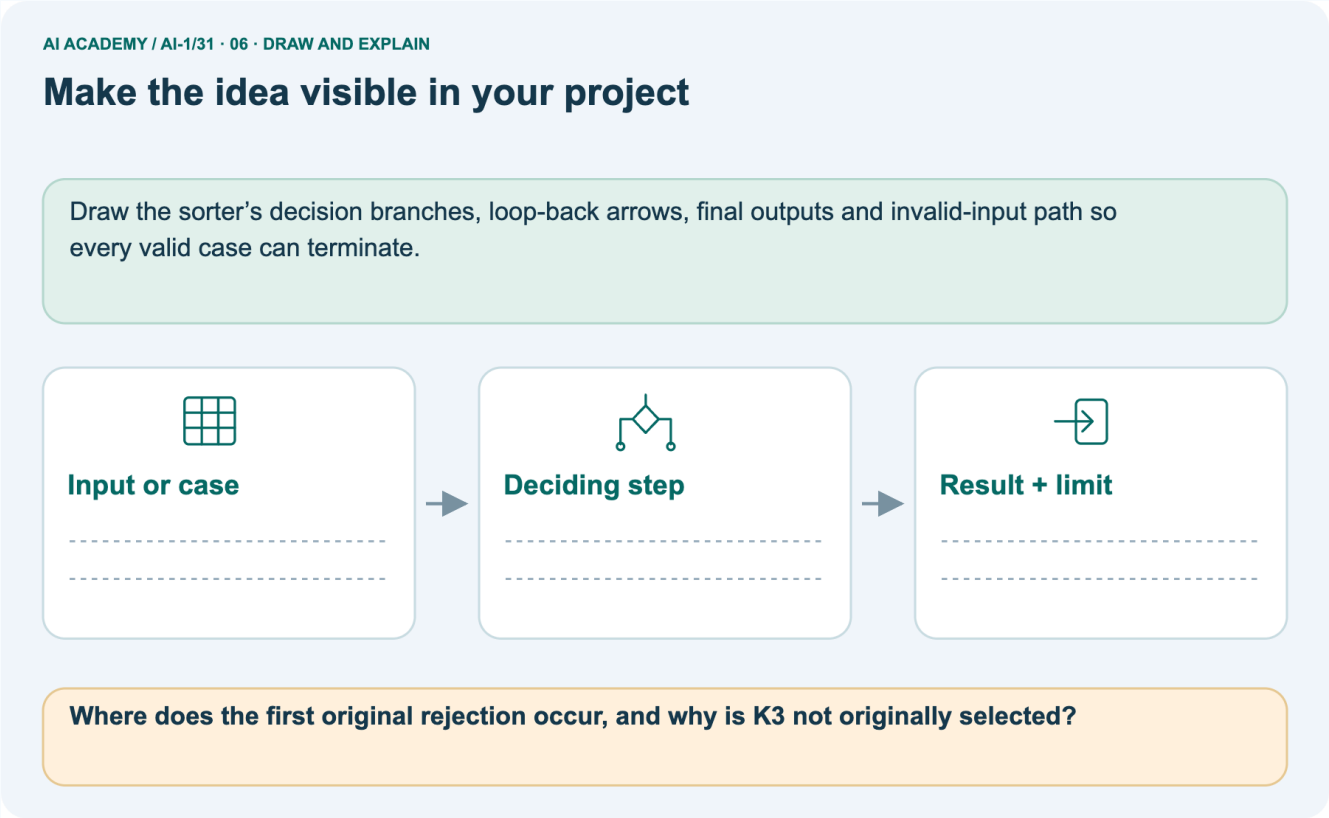
Limit: A correct flow depends on accurate catalogue entries and the declared requirements; the picture does not verify inventory.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the **state**d scope: A correct flow depends on accurate catalogue entries and the declared requirements; the picture does not verify inventory.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate. Draw your **input**, deciding step, result and limit.

Supported question: Where does the first original rejection occur, and why is K3 not originally selected?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: K1 fails glue. K2 meets both conditions, so the first-match flow stops before considering K3.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Design the Decision Process: the case

02. Follow the deciding steps

03. Four paths through the recommender

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.